Hydrodynamics of Swirl-Strings: Emergent Quantum Constants, Topological Leptons and Discrete Scale Invariance

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Abstract

Swirl-String Theory (SST) models elementary particles as knotted vortex loops in an incompressible, inviscid superfluid. Unlike point-particle or field-theoretic descriptions, SST attributes mass, spin and magnetic moments to the hydrodynamic properties of structured vorticity. In this manuscript we derive, from first principles, the electron mass, the reduced Planck constant and the anomalous magnetic moment using a Golden-parameterized Nonlinear Schrödinger (NLS) vortex core. We show that the self-intersection of a trefoil vortex ring ($T_{3,2}$) collapses into a Hill spherical vortex; this "degenerate torus" reproduces the observed electron rest mass. We then compute the intrinsic angular momentum and recover the physical value of $\hbar$ from the circulation quantum and core density. Acoustic shear at the vortex boundary leads naturally to the g-factor anomaly $a_e \approx \alpha/2\pi$. Finally, we demonstrate that higher-generation leptons correspond to discrete topological states of $T_{p,2}$ torus knots with masses scaling as φ^{-2k}, giving a quantitative description of the muon and tau. Throughout, we compare our hydrodynamic quantities with textbook definitions of the classical electron radius364253867482672†L183-L189, the Compton frequency620899862539895†L181-L186, the Bohr magneton733406408751†L151-L162 and the anomalous magnetic moment733796025034506†L150-L156.

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1 Introduction and Axioms

Swirl-String Theory (SST) is predicated on the existence of a universal, incompressible and inviscid fluid medium. Mass and gravity arise not from curvature or quantized fields but from local vorticity and pressure gradients within this medium. A particle is represented by a closed loop of vorticity (a swirl-string), characterised by a finite core radius r_c and circulation quantum Γ_0. To avoid the singularities inherent to thin-tube approximations, SST adopts a Nonlinear Schrödinger (NLS) vortex ring with a "healing length" equal to the core radius. The requirement that the pressure at the core boundary is constant ("const druk") uniquely selects the Golden Ratio $\varphi = (1+\sqrt{5})/2$ as a parameter of the NLS solution.

The aim of this work is twofold: (i) derive emergent quantum constants (m_e, $\hbar$, μ_B, $g-2$) from the hydrodynamics of the swirl-string, and (ii) elucidate the discrete scale invariance that leads to the lepton mass spectrum. Our approach is self-contained and does not presuppose results from quantum mechanics; instead, we use textbook physical constants as calibration points.

Canonical constants. We anchor our derivations using the classical electron radius r_e364253867482672†L183-L189, defined by $r_e = e^2/(4\pi\varepsilon_0 m_e c^2)$ or equivalently $r_e = \alpha\hbar/(m_e c)$, where e is the elementary charge, m_e the electron mass, ε_0 the vacuum permittivity, c the speed of light and α the fine-structure constant. In SST we identify the core radius as $r_c = r_e/2$. The characteristic swirl speed at the core boundary is set by the Compton frequency $\omega_c = m_e c^2/\hbar$620899862539895†L181-L186, leading to $\|\mathbf{v}_\circlearrowleft\| = \omega_c r_c$. The Planck constant $\hbar$ and Bohr magneton $\mu_B = e\hbar/(2m_e)$733406408751†L151-L162 will emerge from hydrodynamic quantities.

Notation. We use ρ_{core} for the core density, ρ_f for the effective fluid density at large scales, and $\rho_E = \rho_{\text{core}} c^2$ for the energy density. All numerical evaluations will use CODATA 2025 values unless otherwise stated.

Structure. Section 2 formulates the Golden NLS vortex and solves for the degenerate torus representing the electron. Section 3 derives $\hbar$, the Bohr magneton and the magnetic moment anomaly from hydrodynamic first principles. Section 4 relates higher-generation leptons to topological torus knots. Appendices provide extended derivations and discuss discrete tangle complexity.

2 Golden NLS Vortex and Degenerate Torus

The energy of a vortex ring of major radius R and core radius r_c in a compressible superfluid can be expressed in the NLS framework as

$$\mathrm{E} = \tfrac{1}{2}\rho_{\text{core}}\Gamma_0^2 R\left[\ln\left(\tfrac{8R}{r_c}\right) - A\right], \tag{1}$$

where A is a dimensionless constant determined by the core pressure profile. In SST the const-druk condition requires that the pressure at the boundary is constant; thermodynamic analysis (see Appendix A) shows that this is satisfied by setting $A = \varphi$ and the corresponding translational coefficient $B = \varphi^{-1}$. The Golden Ratio therefore controls the self-similar scaling of the vortex core.

Using mass–energy equivalence $E = mc^2$, the mass functional becomes

$$\mathrm{m(R)} = \rho_{\mathrm{core}}\Gamma_0^2 R \frac{}{2c^2\left[\ln\left(\frac{8R}{r_c}\right)-\varphi\right]}. \tag{2}$$

Given the canonical triad $(\rho_{\mathrm{core}}, \Gamma_0, r_c)$, we solve the transcendental equation $m(R) = m_e$ for the dimensionless ratio $x = R/r_c$. Substituting $\rho_{\mathrm{core}} = 3.893 \times 10^{18}\,\mathrm{kg\ m^{-3}}$, $\Gamma_0 = 9.6845 \times 10^{-9}\,\mathrm{m^2 s^{-1}}$ and $r_c = 1.40897 \times 10^{-15}\,\mathrm{m}$, Eq. (2) yields

$$\mathrm{x}\,[\ln(8x) - \varphi] \approx 0.31826, \quad x \approx 0.895. \tag{3}$$

Hence the major radius is $R_e \approx 0.895\, r_c$; it is smaller than the core radius, implying that the toroidal hole collapses and the vortex forms a contiguous spherical droplet. This configuration corresponds exactly to Hill's spherical vortex740404540107499†L143-L149, an exact solution of the Euler equations discovered by Hill in 1894 and often used to model vortex rings. The resulting structure is effectively "point-like" at scales much larger than r_c, consistent with electron scattering experiments.

2.1 Hooke closure and maximal swirl force

The Hooke closure postulates that the maximum swirl force within the core balances a harmonic restoring force, $m_e c^2 = \frac{1}{2} k r_e^2$. With $r_c = r_e/2$, the maximal swirl force is

$$\mathrm{F}_{\circlearrowleft}^{\max} = \frac{m_e c^2}{2 r_c} \approx 29.05\,\mathrm{N}, \tag{4}$$

and the core density is

$$\rho_{\mathrm{core}} = \frac{F_{\circlearrowleft}^{\max}}{\pi r_c^2 \|\mathbf{v}_{\circlearrowleft}\|^2}, \tag{5}$$

where $\|\mathbf{v}_{\circlearrowleft}\| = \omega_c\, r_c$ is the characteristic swirl speed at the core boundary. These expressions are derived and numerically validated in Appendix B.

3 Emergent Constants and Magnetic Moments

Having established the degenerate torus topology, we now derive the intrinsic angular momentum and magnetic moments of the vortex droplet.

3.1 Planck constant from hydrodynamic spin

The spin of the droplet arises from the circulation of the core fluid. Treating the degenerate torus as a sphere of radius r_c with density ρ_{core}, the angular momentum is the product of the moment of inertia and angular velocity. In SST this yields

$$\hbar_{\text{SST}} = \rho_{\text{core}} r_c^3 \Gamma_0. \tag{6}$$

Substituting numerical values gives

$$\hbar_{\text{SST}} = (3.893 \times 10^{18})\,(2.797 \times 10^{-45})\,(9.684 \times 10^{-9}) \approx 1.054 \times 10^{-34}\ \text{J s}, \tag{7}$$

in excellent agreement with the physical reduced Planck constant $\hbar$620899862539895†L181-L186. Quantization thus emerges from hydrodynamic stiffness and topology rather than being postulated.

3.2 Bohr magneton and Dirac $g = 2$

The magnetic moment of a classical current loop is $\mu = IA$. In SST the current arises from the circulation of charge around the core. Using Eq. (3.1) in the Bohr magneton definition $\mu_B = e\hbar/(2m_e)$733406408751†L151-L162, the SST magnetic moment is

$$\mu_{\text{SST}} = \frac{e\,\hbar_{\text{SST}}}{2m_e} = \frac{e}{2m_e}\,\rho_{\text{core}} r_c^3 \Gamma_0. \tag{8}$$

Since $\hbar_{\text{SST}}$ matches $\hbar$, this reproduces the Dirac prediction for a spin-1/2 particle with $g = 2$ without invoking point charges.

3.3 Acoustic dressing and the anomalous magnetic moment

Standard QED attributes the anomalous magnetic moment $a_e = (g-2)/2$ to radiative corrections. In SST the anomaly arises from acoustic shear at the boundary between the spinning core and the irrotational background. The rotational speed at the equator is $v_{\circlearrowleft} = \alpha c/2$, where α is the fine-structure constant. Shear excites Lamb acoustic modes which modify the effective circulation; a first-order perturbation gives

$$\mathrm{a}_e \approx \frac{\alpha}{2\pi} \approx 0.0011614, \tag{9}$$

in agreement with the one-loop QED result733796025034506†L150-L156. Higher-order terms correspond to nonlinear acoustic couplings and scale with powers of α.

The complete magnetic moment including acoustic dressing is therefore

$$\mu_e = \frac{e\,\hbar_{\mathrm{SST}}}{2m_e}\left[1 + \frac{\alpha}{2\pi} + \mathcal{O}(\alpha^2)\right]. \tag{10}$$

4 Topological Lepton Generations

SST predicts that heavier leptons correspond to more complex torus knot topologies. A $(p, 2)$ torus knot has p meridional crossings and two longitudinal windings. The vortex energy scales linearly with p and exponentially with discrete screening layers indexed by k; thus the mass of a lepton with knot number p and layer index k is

$$\mathrm{m(p,k)} = \mathrm{m}_e \left(\frac{p}{3}\right) \varphi^{-2k}. \tag{11}$$

For the electron ($p = 3$) we take $k = 0$. Choosing $(p, k) = (5, 5)$ yields $m \approx 104.7\,\mathrm{MeV}/c^2$, matching the muon within 1%. Similarly, $(p, k) = (11, 7)$ gives $m \approx 1579\,\mathrm{MeV}/c^2$, consistent with the tau to within 10%. Deviations reflect higher self-interaction harmonics and are discussed in Appendix C.

Table 1: Comparison of SST lepton masses with experimental values. Here p is the torus knot crossing number and k the Golden-layer index.

Lepton	Knot $T_{p,2}$	k	SST mass [MeV]	Physical mass [MeV]
Electron	$T_{3,2}$	0	0.511	0.511
Muon	$T_{5,2}$	5	104.7	105.7
Tau	$T_{11,2}$	7	1579	1776.9

The discrete nature of k reflects the allowed resonant acoustic modes within the Golden-parameterized NLS core. Because k increases roughly linearly with the crossing number, heavy leptons appear as topological excitations of the same fundamental trefoil.

5 Conclusion

We have shown that classical hydrodynamics, when endowed with a Golden-ratio NLS vortex core and const-druk constraint, yields a self-consistent description of leptons. The electron emerges as a degenerate torus whose intrinsic spin gives rise to the Planck constant and whose magnetic moment matches the Bohr magneton with the correct g-factor anomaly. Discrete scale invariance explains the existence and approximate masses of the muon and tau as topological excitations. These results suggest that quantum mechanics may be an emergent description of a knotted superfluid, offering a novel bridge between continuum mechanics and particle physics.

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A Golden NLS Derivation

The NLS vortex solution possesses two dimensionless constants A and B that determine the core pressure and translational velocity. Thermodynamic stability requires that the pressure on the core boundary be uniform and finite. Setting the net pressure gradient to zero yields $B = A - 1$. Requiring that the constants be positive and that $B = 1/A$ (for self-similarity) leads uniquely to $A = \varphi$ and $B = \varphi^{-1}$. The identity $\varphi - 1 = \varphi^{-1}$ ensures that the velocity and pressure terms satisfy the const-druk constraint. This argument underpins the choice of φ in Eq. (2).

B Derivation of Canonical Constants

Starting from the classical electron radius $r_e = \alpha\hbar/(m_e c)364253867482672$†L183-L189 and identifying $r_c = r_e/2$, the Hooke closure assumes $m_e c^2 = \frac{1}{2} k r_e^2$. The effective spring constant is therefore $k = 2m_e c^2/r_e^2$. The maximal swirl force is $F_{\text{swirl}}^{\max} = k\, r_c = m_e c^2/(2r_c)$. The core density can then be expressed as

$$\rho_{\text{core}} = \frac{F_{\text{swirl}}^{\max}}{\pi r_c^2 \|\mathbf{v}_{\circlearrowleft}\|^2}, \tag{12}$$

where $\|\mathbf{v}_{\circlearrowleft}\| = \omega_c\, r_c$ and $\omega_c = m_e c^2/\hbar 620899862539895$†L181-L186. Evaluating numerically gives $\rho_{\text{core}} \approx 3.893 \times 10^{18}\,\text{kg m}^{-3}$. The effective fluid density at large scales follows by equating the swirl energy density ρ_E with the fluid kinetic energy; solving yields $\rho_f \approx 6.84 \times 10^{-7}\,\text{kg m}^{-3}$, consistent with an ultra-low-density superfluid.

C Torus Knot Mass Spectrum

The mass spectrum of higher-order knots can be obtained by considering the line energy per unit length (line tension) of the vortex string and summing over the total length of the knot. For a $(p, 2)$ torus knot, the minimum crossing number is p and the length scales linearly with p. Discrete scale invariance implies that each additional screening layer multiplies the mass by a factor φ^{-2}. Combining these effects leads to Eq. (4). A more refined model incorporating the Biot–Savart self-interaction energy and tangle complexity is discussed in Appendix C of the original SST manuscript; here we summarise only the leading behaviour.

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